

lab_17

Alexis Saldivar (PID: A69038083)

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)
```

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes

```
data <- read.table("results.txt")
data %>%
  group_by(geno) %>%
  summarize(
    total = n(),
    median = median(exp)
  )
```

```
# A tibble: 3 x 3
  geno total median
  <chr> <int> <dbl>
1 A/A    108  31.2
2 A/G    233  25.1
3 G/G    121  20.1
```

Q14. Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

The SNP affects the expression of ORMDL3. There is a large difference in the median expression values between A/A and G/G genotypes.

```
ggplot(data) +
  aes(x=geno, y=exp, fill=geno) +
  geom_boxplot() +
  geom_jitter(width = 0.2, alpha = 0.3)
```

